COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

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Technology Driven by Innovation

MODULE 1

Data Manipulation

MODULE 1.2

Number System – Arithmetic Operation

At the end of the lesson, the student should be able to:

- Perform Addition, Subtraction, Multiplication and Division of binary, octal and hexadecimal

Decimal Number (Base 10)

- Similar to the ordinary counting numbers from 0 to 9.
- Arithmetic operations such as: Addition, Subtraction, Multiplication and Division are done of the same manner/procedure with performing mathematical operations on counting numbers.

Binary Number (Base 2)

- Binary numbers represents only two numbers , (0 and 1).
- There are different rules in performing arithmetic operations on binary numbers.0

Binary Number (Base 2)

Binary Addition:

0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 0 carry 1

Example:

```
    1011101
+    1101100
    11001001
```

Binary Number (Base 2)

Binary Subtraction:

0 – 0 = 0
0 – 1 = 1 borrow 1
1 – 0 = 1
1 – 1 = 0

Example:

```
   1001101
-  1101000
   -------
   1100101
```

Binary Number (Base 2)

Binary Multiplication:

0 * 0 = 0
0 * 1 = 0
1 * 0 = 0
1 * 1 = 0

Example:

```
     1001101
*        110
  ---------
     0000000
    1001101
   1001101
  ---------
   111001110
```

Binary Number (Base 2)

Binary Division:

- Check the number of digits of the divisor; eg. 3, then take the first 3 digits of the dividend.
- f the dividend is greater than the divisor, then the answer is 1, otherwise the answer is 0.
- Proceed with the process of subtraction, bringing down the next bit of the dividend. Remember you are subtracting binary numbers. Then repeat the process.

Binary Number (Base 2)

Binary Division:

Example:

1001101 / 110

100 / 110 = 0 , 100 – 000 = 100

1001 / 110 = 1, 1001 – 110 = 0011, bring down 1 00111 / 110 = 1,

111 – 110 = 001, bring down 0

0010 / 110 = 0, 0010 – 000 = 10, bring down 1

101 / 110 = 0, 101 – 000 = 101

answer: 1100 r. 101

Hexadecimal Number (Base 16)

- Hexadecimal numbers represents numbers from (0 to 15).
- However, numbers from 10 to 15 are represent by letters for A to F respectively.

- There different rules in performing arithmetic operations on hexadecimal numbers.

Hexadecimal Number (Base 16)

Hexadecimal Addition:

In adding hexadecimal numbers, if the result is greater than 15, it should be subtracted by 16 until the result is less than 16. The number of times you subtracted 16, will be the number of carry on the next bit.

Hexadecimal Number (Base 16)

Hexadecimal Addition:

Example:

a)

```
   B34A
+  1736
  ----
   CA80
```

b)

```
   F6E2
+  5974
  -----
  15056
```

Hexadecimal Number (Base 16)

Hexadecimal Subtraction:

In subtracting hexadecimal numbers, if the number cannot be subtracted then it needs to borrow bit from the next bit, every bit borrowed will be equivalent to 16 and will be added to the bit that needs it, then the process of subtraction will be done.

Hexadecimal Number (Base 16)

Hexadecimal Subtraction:

Example:

a)

```
   2A45
 - F73B
 -------
   5C7A
```

330A

b)

```
   E632
 -
 -------
```

89B8

Hexadecimal Number (Base 16)

Hexadecimal Multiplication:

In multiplying hexadecimal numbers, it also follows the procedure in multiplying decimal numbers, every bit in the multiplier will be multiplied to the multiplicand, if the answer is greater than 15, then subtract 16, until the answer is less than 16. The number of times you subtracted 16 is the number to be carried to the next bit. Answers are arranged just like in multiplying decimal numbers, to obtain the final answer, proceed with the addition process, however remember that you are adding hexadecimal numbers.

Hexadecimal Number (Base 16)

Hexadecimal Multiplication:

Example:

a)

```
      B34A
*       36
    433BC
   +219DE
  +33B5E
  25D19C
```

b)

```
      7632
*       74
    D8C8
```

348EA8

Hexadecimal Number (Base 16)

Hexadecimal Division:

- In dividing hexadecimal numbers, it also follows the procedure in dividing decimal numbers, Check the number of digits of the divisor; eg. 3, then take the first 3 digits of the dividend.
- Proceed with the process of subtraction, bringing down the next bit of the dividend. Then repeat the process. Remember that you are subtracting hexadecimal numbers.

Hexadecimal Number (Base 16)

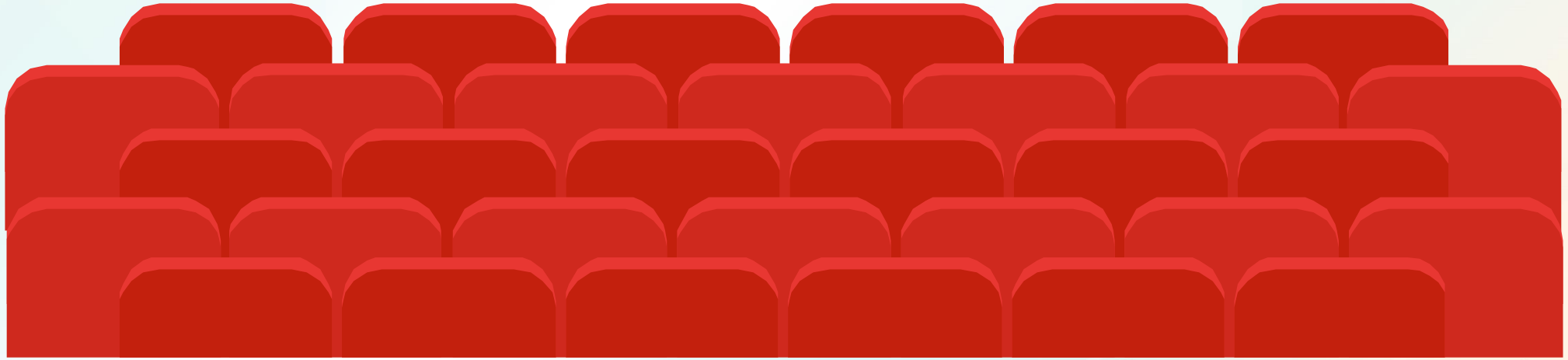
Hexadecimal Division:

Example:

27 / 7 = 5, 7 * 5 in hexadecimal is 23
27 – 23 = 4, 4 cannot be divided by 7

Answer is: 5 remainder 4

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU